Exercise 3: SQL CASE Statement.

1. Table : products

product.id	product.name	price
1	Laptop	1 200
2	Phone	800
3	keyboard	45
4	Monitor	300
5	Mouse	25

Question :

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Expected Outcome Columns
Product.name, price, price.category

```
SELECT product name , price
CASE
    WHEN price > THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
    ELSE 'Budget'
END AS price-category
from products;
```

product.name	price	price.category
Laptop	1 200	Expensive
Phone	800	Mid-Range
keyboard	45	Budget
Monitor	300	Mid-Range
Mouse	25	Budget

2. Table: Orders

Order. id	Customer-name	Amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1 200.00

Question

High-value for orders ≥ 1000.

Medium-value for 500-999.99

'Low Value' for orders < 500

Expected Output Colums:

customer. name

amount

order. value. category

```
SELECT customer_name, amount
CASE
        WHEN amount >= 1 000 THEN 'High Value'
        WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium value'
        ELSE 'Low Value'
END AS order_value_category  from orders;
```

Customer name	Amount	order.value.category
Alice	150.00	Low value
Bob	560.00	Mockum Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

3. Table : employees

emp.id	emp.name	department	salary
1	John	IT	85 000
2	Sara	HR	60 000
3	Mark	IT	75 000
4	Lucy	finance	95 000
5	Tom	HR	55 000

Question:

- If in 'IT' and salary > 8 000 - 'Senior IT'
- If in 'HR' and salary > 5 000 . 'Experienced HR'
- Otherwise -> 'Staff'

Expected Output Columns:
emp.name , department , salary , position.level

```
SELECT emp.name , department , salary,
CASE
        WHEN department ='IT' AND salary > 80 000 THEN 'Senior IT'
        WHEN department = 'HR' AND salary > 55 000 THEN 'Experienced IT'
        ELSE 'Staff'
END AS position.level
from employees;
```

emp name	department	salary	position.level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	~~Staff~~ Experienced HR
Lucy	finance	95 000	Senior IT
Tom	HR	55 000	Staff

Table : Students

Student_id	student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Assign a letter grade:

≥ 90: 'A'	Outcomes:
80-89: 'B'	Student_name
70-79: 'C'	Score
60-69: 'D'	Grade
< 60: 'f'	

```
SELECT student name, score
CASE
    WHEN score >= 90 THEN 'A'
    WHEN score BETWEEN 80 AND 89 THEN 'B'
    WHEN score BETWEEN 70 AND 79 THEN 'C'
    WHEN score BETWEEN 60 AND 69 THEN 'D'
    ELSE 'f'
END AS grade
from students
```

Student_name	Score	Grade
Anna	92	A
Ben	76	C
Cara	59	f
David	83	B
Ella	68	D

5. Table : deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Label delivery performance

- ≤ 30 mins : 'fast'
- 31 - 60 mins : 'On Time'
- 60 mins : Late

Expected Output Columns

- delivery_id
- delivery_time_minutes
- Performance

```
SELECT delivery_id , delivery_time_minutes
CASE
        WHEN delivery_time_minutes <= 30 THEN 'fast'
        WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
        ELSE 'LATE'
END AS performance
FROM deliveries
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	fast
4	65	Late
5	100	Late

6. Table : Tickets

ticket_id	issue type	priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

Convert priority to labels

3 -> 'High'

2 -> 'Medium'

1 -> 'Low'

Expected Output

issue type

priority

priority_~~text~~ label

```
SELECT issue_type, priority,
CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
    END AS priority_label
from tickets;
```

issue_type	priority	priority_level
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Table · Attendance

Student.id	days present	total.days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Each Calculate attendance

- ≥ 90% → 'Excellent'
- 75-89 → 'Good'
- <75% → Needs Improvement

Expected Output

Student.id
attendance_percentage
attendance_status

```
SELECT student.id,
        (days_present * 100 / total_days AS attendance_percentage
CASE
        WHEN (days_present * 100/total_days) ≥ 90= 'Excellent'
        WHEN (days_present * 100/total_days) BETWEEN 75 AND 89 THEN Good
        ELSE 'Needs Improvement'
END AS attendance_status
from attendance;
```

Student.id	attendance_percentage	attendance.status
1	90.0	Excellent
2	60.0	Needs Improvement
3	96.0	Excellent
4	50.0	Needs Improvement
5	100.0	Excellent

8. Table : products_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Label stock status:

- 0 -> Out of Stock'
- 1-5 -> 'low Stock'
- 5 -> 'In Stock'

Expected Output Columns.
product_id
Stock_qty
stock_status

```
SELECT product_id, stock_qty
CASE
        WHEN stock_qty = 0 THEN 'Out of stock'
        WHEN stock_qty BETWEEN 1 AND 4 THEN 'low Stock'
        ELSE 'In Stock'
END AS stock_status
from products_inventory;
```

product_id	stock_qty	stock_status
1	5	In stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9. Table : Classes

Class_id	Subject	Enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Classify by size

- ≥ 25 → 'Large'
- 10 - 24 → 'Medium'
- < 10 → 'Small'

Expected Output Columns

Subject
enrolled_students
class_size_category.

```
SELECT subject,
       enrolled_students
CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    ELSE 'Small'
END AS class_size_category
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10 Table: payments

Payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	Paypal
5	300.00	Cash

Apply discount flag:

- If payment_method = 'Cash' and amount ≥ 200 → 'Eligible for Discount'
- Otherwise → 'Not Eligible'

Expected Output Columns:

- Payment_id
- Payment_method
- amount
- discount_eligibility

```
SELECT payment_id, payment_method, amount,
CASE
    WHEN payment_method = 'Cash' AND amount >= 200 THEN Eligible for
        discount'
    ELSE 'Not Eligible'
END AS discount_eligibility
from payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for discount
3	Card	150.00	Not Eligible
4	Paypal	25.00	Not Eligible
5	Cash	300.00	Eligible for discount